

Business Problem Document

Customer Shopping Behavior Analysis

1. Background

A retail business collects transactional data on every customer purchase, including demographics, product category, payment method, discount usage, subscription status, and customer review ratings. However, this raw data on its own does not tell the business which customers, products, or channels are actually driving performance. Without structured analysis, decisions about marketing spend, discounting strategy, and customer retention are being made without clear evidence.

2. Business Problem

The business needs to understand: which customer segments generate the most revenue, whether subscribed customers behave differently from non-subscribers, which products and categories are top performers, how discounting affects purchase behavior, and how customers can be segmented by loyalty for targeted retention efforts. Answering these questions requires moving from raw transactional data to a structured analysis that can inform real decisions.

3. Objectives

- Clean and prepare the raw customer transaction dataset for analysis
- Load the cleaned data into a relational database (MySQL) for structured querying
- Answer specific business questions using SQL, covering revenue, customer segments, product performance, and discount behavior
- Build an interactive Power BI dashboard to visualize the findings for a non-technical audience
- Deliver a written report and presentation summarizing key insights and recommendations

4. Data Source

Kaggle "Customer Shopping Trends Dataset" (approximately 3,900 records, 18 columns), covering customer demographics, purchase details, product categories, payment methods, subscription status, and review ratings. This is a publicly available dataset used for portfolio and educational purposes, not real company data.

5. Scope

This project covers data cleaning and exploration (Python), business-question analysis (SQL), dashboard visualization (Power BI), and a final written report with recommendations. It does not include predictive modeling or real-time data integration.